

Answer of Question No. 6

a) let, $\bar{x}$ and $\bar{y}$ denote the means of X and Y respectively, i.e., $E(X) = \bar{x}$ and $E(Y) = \bar{y}$.

$$E(X - \bar{x}) = \sigma_x^2 \text{ and } E(Y - \bar{y}) = \sigma_y^2.$$

If we write $U = X + Y$. we have to find the correlation coefficient between X and U.

The mean of U is $E(U) = E(X + Y) = E(X) + E(Y)$

$$\text{Or, } \bar{u} = \bar{x} + \bar{y}$$

$$\text{Hence, } U - \bar{u} = (X + Y) - (\bar{x} + \bar{y}) = (X - \bar{x}) + (Y - \bar{y})$$

$$\text{Var}(U) = \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2.\text{Cov}(X, Y)$$

$$= \sigma_x^2 + \sigma_y^2, \text{ as X and Y are independent, as } \text{Cov}(X, Y) = 0.$$

$$\text{i.e., } \sigma_U = \sqrt{\sigma_x^2 + \sigma_y^2} \quad [2 \text{ marks}]$$

$$\text{Again, } \text{Cov}(X, U) = E\{(X - \bar{x}). (U - \bar{u})\}$$

$$= E[(X - \bar{x})\{(X - \bar{x}) + (Y - \bar{y})\}]$$

$$= E(X - \bar{x})^2 + E\{(X - \bar{x}). (Y - \bar{y})\}$$

$$= \sigma_x^2 + \text{Cov}(X, Y) = \sigma_x^2 \text{ as } \text{Cov}(X, Y) = 0. \quad [2 \text{ marks}]$$

Hence, the required correlation coefficient is

$$\rho_{X, (X+Y)} = \frac{\text{Cov}(X, U)}{\sigma_X \sigma_U} = \frac{\sigma_x^2}{\sigma_x \cdot \sqrt{\sigma_x^2 + \sigma_y^2}} = \frac{\sigma_x}{\sqrt{\sigma_x^2 + \sigma_y^2}} \quad [1 \text{ mark}]$$

b) Let, (X, Y) and (U, V) are two sets of random variables are related with a relationship that

$U = \frac{X - c}{d}$ and $V = \frac{Y - c'}{d'}$, where c, d, c', d' are constant. Again, $\bar{x}, \bar{y}, \bar{u}, \bar{v}$ are means of random variables X, Y, U, and V respectively.

$$\text{Now, } \text{Var}(U) = E(U - \bar{u})^2 = E\left(\frac{X - c}{d} - \frac{\bar{x} - c}{d}\right)^2 = \frac{\text{Var}(X)}{d^2} \rightarrow \sigma_x^2 = d^2 \cdot \sigma_U^2$$

$$\text{Similarly, } \sigma_y^2 = d'^2 \cdot \sigma_V^2$$

$$\text{Cov}(U, V) = E\{(U - \bar{u}). (V - \bar{v})\} = E\left[\left(\frac{X - c}{d} - \frac{\bar{x} - c}{d}\right) \cdot \left(\frac{Y - c'}{d'} - \frac{\bar{y} - c'}{d'}\right)\right]$$

$$= \frac{1}{dd'} E\{(X - \bar{x}). (Y - \bar{y})\} = \frac{1}{dd'} \cdot \text{Cov}(X, Y)$$

$$\text{Therefore, } b_{yx} = \frac{\text{Cov}(X, Y)}{\sigma_x^2} = \frac{dd' \text{Cov}(U, V)}{d^2 \cdot \sigma_U^2} = \frac{d'}{d} \cdot b_{vu} \quad [1 \text{ mark}]$$

$$\text{Similarly, } b_{xy} = \frac{d}{d'} \cdot b_{uv}$$

The regression coefficients depend on d and d' not on c and c'. Therefore, it is invariant under origin change but affected by scale change. [1 mark]

c) Given, Regression line of y on x: $8x - 10y + 66 = 0 \dots (1)$

Regression line of x on y: $40x - 18y = 214 \dots (2)$

At any rate, the regression line always passes through the means of X and Y. We have,

$$8\bar{x} - 10\bar{y} + 66 = 0 \dots (3)$$

$$40\bar{x} - 18\bar{y} - 214 = 0 \dots (4)$$

From (3), $\bar{x} = \frac{1}{8}(10\bar{y} - 66)$ and from (4), $\bar{x} = \frac{1}{40}(18\bar{y} + 214)$

Equating these two values,

$$\frac{1}{8}(10\bar{y} - 66) = \frac{1}{40}(18\bar{y} + 214)$$

$$\Rightarrow 10\bar{y} - 66 = \frac{1}{5}(18\bar{y} + 214)$$

$$\Rightarrow 50\bar{y} - 330 = 18\bar{y} + 214$$

$$\Rightarrow 32\bar{y} = 544$$

$$\Rightarrow \bar{y} = 17$$

Putting this value in (3), we get,

$$8\bar{x} - 10 \cdot 17 + 66 = 0$$

$$\Rightarrow 8\bar{x} = 104$$

$$\Rightarrow \bar{x} = 13$$

Thus, we have, mean of x = 13 and mean of y = 17. [2 marks]

Again, from equation (1), we have:

$$y = \frac{8}{10}x + \frac{66}{10}$$

$$\Rightarrow y - 17 = \frac{8}{10}(x - 13) + \frac{66}{10} - 17 + \frac{8}{10} \cdot 13$$

$$\Rightarrow y - 17 = \frac{8}{10}(x - 13)$$

Comparing this equation with the equation of line of regression of y on x, i.e.,

$$y - \bar{y} = b_{yx}(x - \bar{x}), \text{ we have, } b_{yx} = \frac{8}{10} = \frac{4}{5}$$

Again, from equation (2), we have,

$$\begin{aligned}
x &= \frac{18}{40}y + \frac{214}{40} \\
\Rightarrow x - 13 &= \frac{18}{40}(y - 17) + \frac{214}{40} - 13 + \frac{18}{40} * 17 \\
\Rightarrow x - 13 &= \frac{18}{40}(y - 17) + \frac{1}{40}(214 - 520 + 306) \\
\Rightarrow x - 13 &= \frac{18}{40}(y - 17)
\end{aligned}$$

Comparing this equation with the equation of line of regression of x on y, i.e.,

$$x - \bar{x} = b_{xy}(y - \bar{y}), \text{ we have, } b_{xy} = \frac{18}{40} = \frac{9}{20}$$

Also, we know, $r^2 = b_{yx} * b_{xy}$. Hence,

$$\begin{aligned}
r^2 &= b_{yx} * b_{xy} = \frac{4}{5} * \frac{9}{20} = \frac{9}{25} \\
\Rightarrow r &= \pm \frac{3}{5} = \pm 0.6
\end{aligned}$$

Therefore, the value of correlation coefficient is $r = 0.6$. [2 marks]

Again, given that, $\sigma_x^2 = 9$. Therefore, regression coefficient,

$$b_{yx} = r * \frac{\sigma_y}{\sigma_x} \Rightarrow b_{yx}^2 = r^2 * \frac{\sigma_y^2}{\sigma_x^2} \Rightarrow \frac{16}{25} = \frac{9}{25} * \frac{\sigma_y^2}{9} \Rightarrow \sigma_y^2 = 16.$$

Hence, $\sigma_y^2 = 16$. [1 mark]